

HPC Assignment 03: Serial Interpolation on Scattered Data

Aalok Thakkar

Roll Number: 202301429

Khushi Pandya

Roll Number: 202301406

Dhirubhai Ambani University

February 24, 2026

Abstract

This report presents the implementation and performance analysis of a serial bilinear interpolation algorithm for mapping scattered 2D data points onto a structured mesh grid. The algorithm distributes particle values to four nearest grid points using area-based weighting. Performance is evaluated on both Lab PC and HPC cluster across five different problem configurations, with detailed analysis of time complexity, memory access patterns, and cache behavior.

1 Implementation Approach

1.1 Problem Overview

Given N scattered points $S = \{(x_i, y_i, f_i) | i = 1, 2, \dots, N\}$ where $f_i = 1$, we interpolate these values onto a structured $M \times M$ mesh grid $G = \{(X_i, Y_j)\}$ using bilinear interpolation.

1.2 Grid Structure and Cell Identification

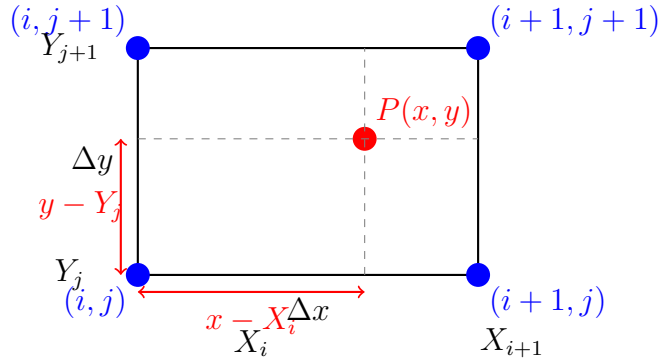


Figure 1: Grid cell structure showing a particle P at position (x, y) within cell (i, j)

1.2 Grid Structure and Cell Identification

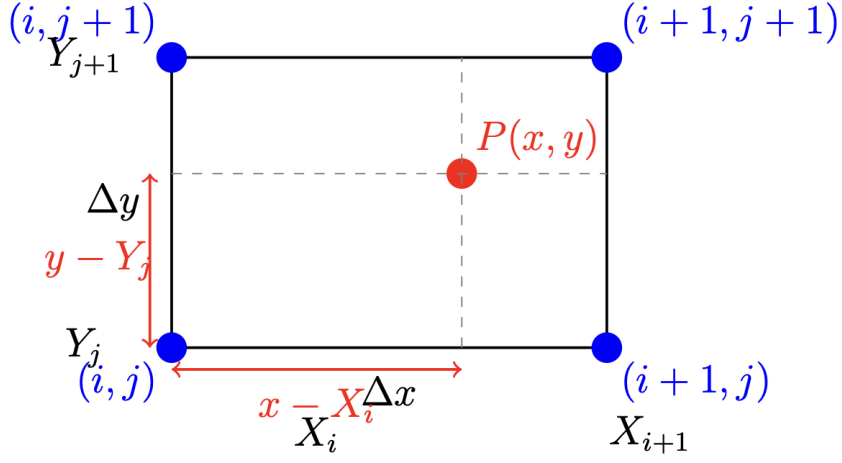


Figure 1: Grid cell structure showing a particle $P(x, y)$ within cell (i, j) .

For each scattered point (x, y) :

1. Cell Identification

$$i = \left\lfloor \frac{x}{\Delta x} \right\rfloor, \quad j = \left\lfloor \frac{y}{\Delta y} \right\rfloor \quad (1)$$

where

$$\Delta x = \frac{1}{N_x}, \quad \Delta y = \frac{1}{N_y}$$

2. Normalized Distances

$$dx = \frac{x - X_i}{\Delta x}, \quad dy = \frac{y - Y_j}{\Delta y} \quad (2)$$

3. Clamp to Valid Range

If $i \geq N_x$, set $i = N_x - 1$ (similarly for j).

1.3 Weight Calculation (Area-Based)

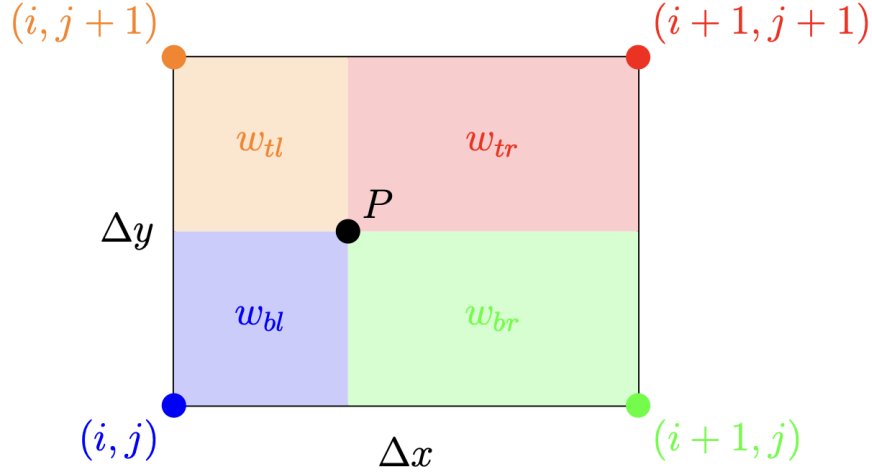


Figure 2: Area-based weight calculation. Each corner receives weight proportional to the opposite sub-rectangle area.

$$w_{i,j} = (1 - dx)(1 - dy) \quad (3)$$

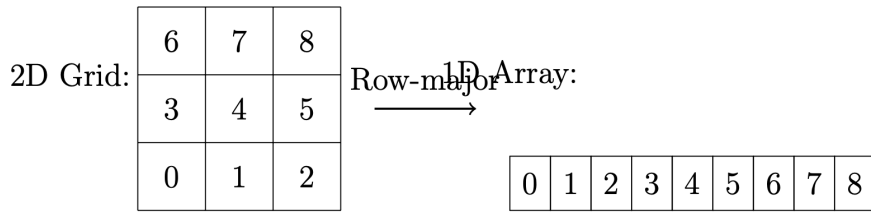
$$w_{i+1,j} = dx(1 - dy) \quad (4)$$

$$w_{i,j+1} = (1 - dx)dy \quad (5)$$

$$w_{i+1,j+1} = dxdy \quad (6)$$

$$\sum w = 1$$

1.4 Memory Layout



$$\text{Index formula: } \text{idx} = j \times \text{GRID_X} + i$$

$$\text{Example: } \text{idx}(1,2) = 2 \times 3 + 1 = 7$$

Figure 3: Row-major memory layout for storing the 2D mesh in a 1D array.

The mesh is stored in row-major order:

$$\text{mesh_value}[j \times \text{GRID_X} + i] = F(X_i, Y_j) \quad (7)$$

2 Theoretical Analysis

2.1 Time Complexity

$$O(\text{Maxiter} \times \text{Npoints})$$

3 Performance Results

3.1 Performance Visualizations

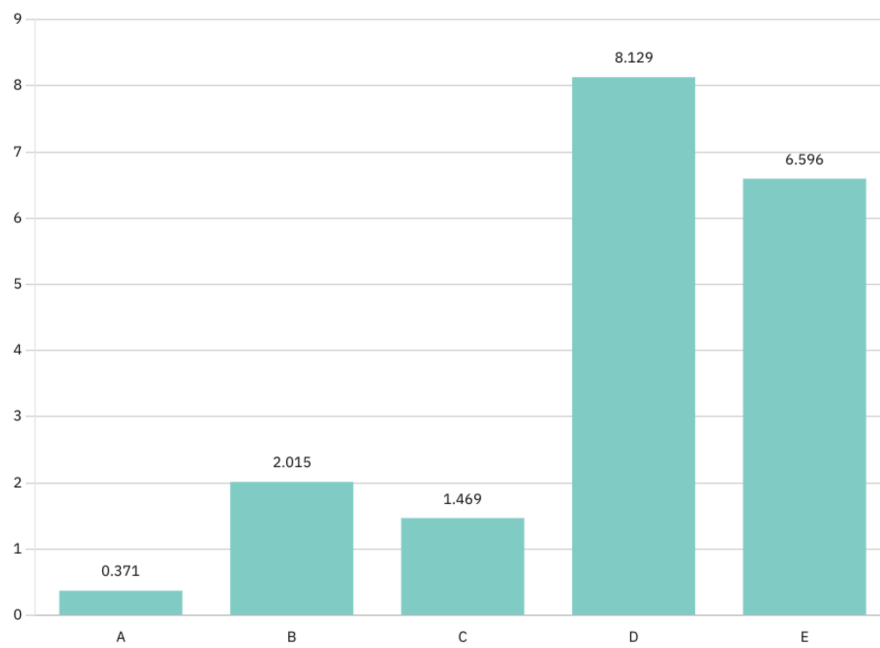


Figure 4: Execution time on Lab PC across test configurations.

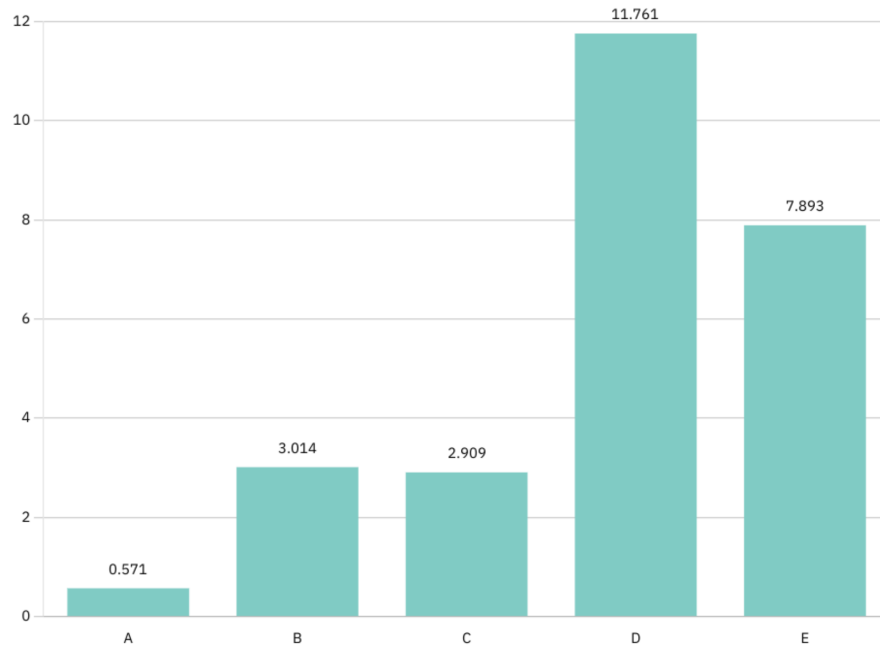


Figure 5: Execution time on HPC Cluster across test configurations.

4 Conclusion

- Linear time complexity.
- Memory-bound performance due to scattered writes.
- Serial performance strongly depends on single-core characteristics.